Problem Set 4

Firms, Production, and Production Functions

EC306: Intermediate Microeconomics

Instructions. Work through these in class. Show all working and interpret your numerical answers in words. Worked solutions follow at the end.

Question 1 — Product curves

A firm's short-run production function is $q = 18L^2 - L^3$ (K fixed).

(a) Derive MP_L and AP_L.

(b) Find the level of L at which (i) MP_L is maximised, (ii) AP_L is maximised, (iii) total product is maximised.

(c) Verify algebraically that $\text{MP}_L = \text{AP}_L$ at the maximum of AP_L, and explain in one sentence why this must always hold.

(d) Beyond which level of L does the law of diminishing marginal returns set in here? Explain what is physically "diminishing" and why it is a statement about *fixed* capital, not about worker quality.

Question 2 — Isoquants, MRTS, returns to scale

Consider $q = AL^\alpha K^\beta$.

(a) Derive MP_L, MP_K, and the MRTS. Show the MRTS is diminishing in L along an isoquant.

(b) Prove the general returns-to-scale result in terms of $\alpha + \beta$.

(c) For the CES form $q = (aL^\rho + bK^\rho)^{1/\rho}$, derive the MRTS and identify the limiting technologies as $\rho \to 1$ and $\rho \to -\infty$.

(d) The elasticity of substitution for CES is $\sigma = 1/(1 - \rho)$. Explain in 3–4 sentences what σ measures, and why the question "how easily does capital substitute for labour?" is central to the current debate about automation and jobs.

Question 3 — Returns to scale in a real industry

(a) Choose a real industry and argue, with evidence or a clear mechanism, whether it exhibits increasing, constant, or decreasing returns to scale. Cite at least one source (industry report, paper, or credible data) on its cost or productivity structure.

(b) Identify the specific source of the scale property (e.g. fixed R&D cost spread over volume, network effects, geometric volume-to-surface relationships, coordination costs).

(c) If your industry has increasing returns to scale, explain in 2–3 sentences what this implies for market structure (why such industries tend toward few large firms). If it has decreasing returns, explain what limits firm size.

Question 4 — Fixed proportions in practice

A logistics firm uses drivers (L) and trucks (K): $q = \min\{L, K\}$ deliveries.

(a) If the firm has $K = 20$ trucks and $L = 26$ drivers, what is output, and how many drivers are idle? What is the MP of a 27th driver?

(b) Explain why the isoquants are L-shaped and why the MRTS is undefined at the corner.

(c) Real haulage is not perfectly Leontief — name two margins on which drivers and trucks *are* substitutable (e.g. shift patterns, vehicle size), and explain how relaxing the fixed-proportions assumption would change the firm's response to a driver shortage.

Solutions

Question 1

(a) $\mathrm{MP}_L = \dfrac{dq}{dL} = 36L - 3L^2$; $\mathrm{AP}_L = \dfrac{q}{L} = 18L - L^2$.

(b) (i) MP_L max: $\dfrac{d\mathrm{MP}_L}{dL} = 36 - 6L = 0 \Rightarrow L = 6$, $\mathrm{MP}_L(6) = 216 - 108 = 108$. (ii) AP_L max: $\dfrac{d\mathrm{AP}_L}{dL} = 18 - 2L = 0 \Rightarrow L = 9$, $\mathrm{AP}_L(9) = 162 - 81 = 81$. (iii) TP max: $\mathrm{MP}_L = 0 \Rightarrow 3L(12 - L) = 0 \Rightarrow L = 12$, $q = 18(144) - 1728 = 864$.

(c) At $L = 9$: $\mathrm{MP}_L(9) = 324 - 243 = 81 = \mathrm{AP}_L(9)$. This always holds because AP_L rises when $\mathrm{MP}_L > \mathrm{AP}_L$ and falls when $\mathrm{MP}_L < \mathrm{AP}_L$; at the maximum of AP_L it is momentarily flat, which forces $\mathrm{MP}_L = \mathrm{AP}_L$.

(d) Diminishing marginal returns begin where MP_L starts to fall: $\dfrac{d\mathrm{MP}_L}{dL} < 0 \Rightarrow L > 6$. What "diminishes" is the *extra* output per extra worker, because the *fixed* capital is spread across more workers (each has less capital to work with). It is a statement about the fixed factor, not about worker quality — workers are identical here.

Question 2

(a) $\mathrm{MP}_L = A\alpha L^{\alpha-1}K^\beta$, $\mathrm{MP}_K = A\beta L^\alpha K^{\beta-1}$, so $\mathrm{MRTS} = \dfrac{\mathrm{MP}_L}{\mathrm{MP}_K} = \dfrac{\alpha}{\beta}\dfrac{K}{L}$. Moving down an isoquant ($L\uparrow, K\downarrow$) lowers K/L, hence MRTS falls — diminishing MRTS.

(b) Scaling inputs by t: $A(tL)^\alpha(tK)^\beta = t^{\alpha+\beta}AL^\alpha K^\beta$. So $\alpha + \beta > 1$: increasing; $= 1$: constant; < 1: decreasing returns to scale.

(c) CES $q = (aL^\rho + bK^\rho)^{1/\rho}$: $\mathrm{MRTS} = \dfrac{aL^{\rho-1}}{bK^{\rho-1}} = \dfrac{a}{b}\Big(\dfrac{K}{L}\Big)^{1-\rho}$. As $\rho \to 1$, $\mathrm{MRTS} \to a/b$ (constant) — *perfect substitutes*, linear isoquants. As $\rho \to -\infty$, the isoquants become L-shaped — *fixed proportions (Leontief)*.

(d) $\sigma = 1/(1-\rho)$ is the elasticity of substitution: the % change in the K/L ratio per % change in the MRTS — how easily capital substitutes for labour. It is central to the automation debate: $\sigma > 1$ means machines readily replace workers (automation displaces labour, capital's share rises), whereas $\sigma < 1$ means capital and labour are complements and automation tends to raise labour demand.

Question 3 (model answer / marking notes)

Data-dependent. **(a)** names a real industry and argues IRS/CRS/DRS with a cited cost-or-productivity source. **(b)** pins the *mechanism*: fixed R&D/fixed cost spread over volume, network effects, or geometric volume-to-surface ratios (IRS) vs. coordination/managerial limits or scarce inputs (DRS). **(c)** IRS $\Rightarrow$ falling average cost favours few large firms (natural concentration); DRS $\Rightarrow$ rising average cost limits firm size and supports many firms.

Question 4

(a) $q = \min\{L, K\}$ with $K = 20, L = 26$: $q = \min\{26, 20\} = 20$. Idle drivers $= 26 - 20 = 6$. The 27th driver's MP $= 0$ (trucks are the binding input).

(b) Isoquants are L-shaped because adding only one input (with the other fixed) yields no extra output; at the corner the isoquant has no unique tangent slope, so the MRTS is undefined.

(c) *Model answer.* Real haulage is not perfectly Leontief — name two substitution margins, e.g. shift patterns/double-manning (more driver-hours per truck) and vehicle size/load (more output per driver). Relaxing fixed proportions gives a finite, downward-sloping MRTS, so a driver shortage can be partly absorbed by working trucks harder or using larger vehicles rather than leaving capital idle.